



Umair Qudus

Researcher – Knowledge Graphs, Verifiable AI & ML for Science

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Marie Skłodowska-Curie Fellow and Ph.D. researcher (thesis submitted, Aug 2026) at Paderborn University's DICE Research Group, specialising in automated fact-checking and verifiable AI over large-scale Knowledge Graphs. Six years of research combining graph traversal, KG embeddings, temporal reasoning, NLP, and LLMs — published across ACM Computing Surveys, ISWC, ESWC, CIKM, EKAW, Semantic Web Journal, and Springer (9 publications). Proficient in Python, PyTorch, HuggingFace, RDF/OWL, SPARQL, GraphDB, Virtuoso, and PyG.

Employment

- 2020–present** **Researcher (Wissenschaftlicher Mitarbeiter) & Ph.D. Candidate**
Paderborn University – DICE Research Group, Germany
EU Horizon 2020 **Marie Skłodowska-Curie Fellow** (KnowGraphs ITN); Paderborn researcher on BMBF-funded **NEBULA** misinformation detection project. Doctoral research on automated KG fact-checking published in premium conferences/journals in semantic web: *HybridFC*[C1] (**ISWC 2022**), *TemporalFC*[C2] (**ISWC 2023**), *ExPrompt*[C3] (**CIKM 2024**), *FAVEL*[C4] (**EKAW 2024**), *Shall-Know*[C5] (**ESWC 2026**), and a RAG pipeline for misinformation detection in NEBULA project (**Springer 2026**). Survey[J2] of KG fact-checking approaches in **ACM Computing Surveys** (2025); ML lecturer (Master's level); reviewer for AAAI, IJCAI, ISWC, ESWC, CIKM, SWJ.
- 2017–2020** **Research Engineer**
Kyung Hee University, South Korea
Designed a DCAT-based dataset recommendation system using ML-similarity over DBpedia. Built a SPARQL-to-API mapping tool translating queries to REST calls and RDF. Developed a federated query processing quality benchmark.
- 2016–2017** **University Lecturer**
SZABIST University, Pakistan
Designed and delivered university-level courses in Machine Learning, Artificial Intelligence, and Algorithms to undergraduate and postgraduate students.
- 2015–2016** **Co-Founder (part time)**
IT Art Technologies, Islamabad, Pakistan
Co-founded a mobile app startup delivering Android and iOS applications for freelance clients and proprietary products, including a published app on Google Play.
- 2013–2015** **Software Engineer**
Convo.com, Islamabad office, Pakistan
Developed and shipped features (real-time chat, news feed, search) for a business collaboration platform serving 6,000+ enterprise clients including TechCrunch and CBS, backed by Morgenthaler Ventures.

Education

- 2020–2026** **Ph.D. in Computer Science**
Paderborn University, Germany
Thesis title: *Fact-Checking over Knowledge Graphs*
(Advisor: Prof. Dr. Axel-Cyrille Ngonga Ngomo)

2014–2016	M.Sc. in Computer Science <i>National University of Computing and Emerging Sciences, Pakistan</i> Thesis: <i>Solving Constraint Satisfaction Problems (CSP) Using Neural Backpropagation</i>
2009–2013	B.Sc. in Computer Science <i>National University of Computing and Emerging Sciences, Pakistan</i> Focus: <i>Computer Science</i> (CGPA: 3.52/4.0)

Technical Skills

Knowledge Graphs & Semantic Web

- RDF, OWL, SPARQL, DCAT, DBpedia
- Knowledge graph embeddings, graph traversal
- Temporal knowledge graphs, KG quality, fact validation
- Federated SPARQL query processing, semantic data integration

Generative AI, NLP & ML

- Large Language Models, prompt engineering, stance classification
- Fact checking, misinformation detection, factuality assessment
- PyTorch, scikit-learn, Pandas, NumPy

Programming & Research Engineering

- Python, Java, SQL, JavaScript, C++
- Docker, Git, Linux, REST APIs
- MySQL, Oracle, Android

Languages

- English – Full professional proficiency
- German – Professional working proficiency
- Urdu – Native
- Korean – Limited working proficiency

Honors & Awards

- EU Horizon 2020 Marie Skłodowska-Curie Fellowship – Paderborn University
- Science Day Competition Winner – Paderborn University ¹
- Best research poster award – Artificial Intelligence and Knowledge Graphs for Scholarly Data – at Leuphana University Lüneburg ²
- Silver Medal & Dean's List of Honor – National University of Computing and Emerging Sciences
- Final Year Project Funded by ICT (Pakistan)
- **Reviewing:** Semantic Web Journal, ISWC, ESWC, AAI, IJCAI, and CIKM

References

Available upon request.

¹<https://cs.upb.de/cs-news-single-view/science-day-2022-nach-zwei-jahren-pause-wieder-da>

²<https://sites.google.com/view/aikg-sd-summer-school-day-2024/startseite>

Awards & Publications

In the following, I list my notable mentions and publications during PhD. All publications went through a peer-review.

Awards and Notable Mentions

- [A1] Best Poster award at Artificial Intelligence and Knowledge Graphs for Scholarly Data–Summer School at Leuphana University Lüneburg 2024.
Umair Qudus, Michael Röder, Axel-Cyrille Ngonga Ngomo: *Fake news detection Using Augmented Prompt Methodologies and KGs*
- [A2] Science Day Competition Winner at Paderborn University 2022.
Umair Qudus: *Fact checking over Knowledge Graphs*

Journal Articles

- [J1] **Umair Qudus**, Michael Röder, Muhammad Saleem, and Axel-Cyrille Ngonga Ngomo: *Fact checking Knowledge Graphs–A survey*, ACM Computing Survey, CSUR 2025.
- [J2] **Umair Qudus**, Muhammad Saleem, and Axel-Cyrille Ngonga Ngomo, Young-koo Lee: *An Empirical Evaluation of Cost-based Federated SPARQL Query Processing Engines*, Semantic Web, 2021.

Conference Articles

- [C1] **Umair Qudus**, Michael Röder, Muhammad Saleem, and Axel-Cyrille Ngonga Ngomo: *HybridFC: A Hybrid Fact-Checking Approach for Knowledge Graphs*, ISWC 2022.
- [C2] **Umair Qudus**, Michael Röder, Sabrina Kirrane, and Axel-Cyrille Ngonga Ngomo: *TEMPORALFC: Temporal Fact Checking Approach over Knowledge Graphs*, ISWC 2023.
- [C3] **Umair Qudus**, Franck Lionel Tatkeu Pekarou, Ana Alexandra Morim da Silva, Michael Röder, and Axel-Cyrille Ngonga Ngomo: *FAVEL: Fact Validation Ensemble Learning*, EKAW 2022.
- [C4] **Umair Qudus**, Michael Röder, Daniel Vollmers, and Axel-Cyrille Ngonga Ngomo: *EXPROMPT: Augmenting Prompts Using Examples as Modern Baseline for Stance Classification*, CIKM 2024.
- [C5] **Umair Qudus**, Neha Pokharel, Michael Röder, and Axel-Cyrille Ngonga Ngomo: *SHALLKNOW: No Need to Be a Know-It-All: Fact Checking with Shallow Knowledge*, ESWC 2026.
- [C6] Abullah Qamar, **Umair Qudus**³, Michael Röder, and Axel-Cyrille Ngonga Ngomo: *RANGEFC: Interval-Aware Temporal Fact Checking for Knowledge Graph*, ISWC 2026.

Book Chapter

- [B1] **Umair Qudus**, Michael Röder, and Axel-Cyrille Ngonga Ngomo: *Automatic Detection of Dis- and Misinformation using Shallow Knowledge Graphs*, Springer 2026.

Poster

- [P1] **Umair Qudus**, Michael Röder, Axel-Cyrille Ngonga Ngomo: *Fake news detection Using Augmented Prompt Methodologies and KGs*, Poster at Artificial Intelligence and Knowledge Graphs for Scholarly Data–Summer School at Leuphana University Lüneburg 2024.

³(equal contribution as first author.)